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Why Eggs Matter for Schoolkids: The Midday Meal Debate

Why eggs matter for schoolkids

Different foods serve distinct nutritional purposes. But from the lens of nutrition science, the egg remains one of the most complete and easy-to-prepare foods



EXPERT EXPLAINS

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AT A time when several states continue to debate whether eggs should be part of school midday meals, West Bengal has pulled them out. The discussion often centres on culture, ideology or personal food choices. Look in that debate is a fundamental question: What does science say?

For millions of Indian children, especially those from low-income households, the midday meal may be their most nutritious meal of the day. Decisions about what goes on that plate must, therefore, be guided by evidence. And the evidence on eggs — as a source of complete, affordable and highly bioavailable protein (which is well-utilised by the body upon consumption) — is difficult to ignore.

All nutritional guidelines suggest that 30-35% of our daily calories should come from protein-rich foods. Protein is also necessary for building lean mass, burning fat, increasing metabolism, slowing blood sugar release, and helping the body repair itself. However, "The issue is not just of protein from animal or plant sources. The most important is how much of that protein can the body actually use,"

Dr. Seema Gulati, Head, Nutrition Research Group at the National Diabetes, Obesity and Cholesterol (N-DOOC) Foundation, Delhi, tells *Khaskhaskhaskh*. N-DOOC is a non-profit that partners with the Indian Council of Medical Research to research and prevent metabolic diseases.

The answer begins not with grams of protein, but with amino acids, the tiny building blocks of life from which every muscle fibre, hormone, enzyme and immune cell is made. The human body requires 20 amino acids, of which 11 can be manufactured internally. The remaining nine must come entirely from food.

A food that supplies all nine in adequate proportions is called a complete protein. Eggs are among the very few natural foods that do precisely that; hence they have been a part of midday meal schemes. Those who cannot have eggs need a mix of plant proteins for equivalence," Gulati adds. Here, she explains why eggs are often seen as the gold standard for protein intake, and what it takes to match their nutritional profile for people who don't consume them.

Protein: Quantity vs quality

Consider the numbers (see box). A hundred grams in terms of real meals means two boiled eggs, a two-thirds cup of cooked soybeans and more than a cup of paneer chunks. In this scenario, cooked soybeans contain 13.2 g of protein, almost identical to the 13.4 g of protein in two boiled eggs. Paneer provides 18.9 g, while whole milk contributes 13.3 g.

Relatively by protein content, eggs and cooked soybeans appear virtually indistinguishable. But protein isn't all that's created equal. Egg protein naturally contains all nine essential amino acids in proportions remarkably close to human requirements. It is also highly digestible, allowing the body to absorb and utilise it more efficiently compared to other foods. This bioavailability is why eggs have long served as the international reference protein against which the quality of other proteins is assessed.

Plant proteins tell a more complex story. Soybeans is undoubtedly among the highest quality plant proteins avail-

able and comes closer than most plant foods to providing a balanced amino acid profile. Yet, like other plant proteins, it's best viewed as part of a dietary mix rather than a standalone solution.

Nutritionists, therefore, recommend combining legumes with cereals — dal with rice, rajma with rice, khichdi, mil with chana, or millets with pulses — so that one food compensates for the amino acids another supplies in smaller amounts. For vegetarians, this principle of protein complementation is food mixed with legumes, dairy and nuts. An egg, by contrast, has that balance already built in. Plant proteins may seem inadequate individually, but they achieve their full nutritional potential through intelligent combinations. Cereals and pulses complement one another. Dairy and soy enrich the overall amino acid profile. Apart from combinations, there are always fortified foods.

The 'brain nutrient' factor

Choline is a nutrient that receives far less public attention than protein, despite being indispensable for brain development, memory, nerve signalling and heart health. Whole milk provides around 14-16 mg of choline per 100 g. Paneer contains roughly 15-20 mg, while cooked soybeans offers about 44-45 mg. Eggs provide 260-300 mg of choline. In simple terms, eggs contain almost seven times more choline than cooked soybeans and nearly 20 times more than milk or paneer.

Each food has its own strength

Each food brings something valuable to the table. Cooked soybeans are an excellent source of dietary fibre (792g), iron (2.8 mg), magnesium (66.1 mg) and potassium (579 mg). It also contains useful amounts of thiamine, niacin and folate, making it one of the strongest plant-based protein foods available. Paneer stands out for its 18.9 g of protein and its exceptionally high calcium content (476 mg per 100 g), making it an important food for bone health. It also contributes phosphorus, zinc and biotin in meaningful amounts.

Milk remains one of the most accessible sources of calcium, riboflavin and high-quality dairy nutrition. A boiled egg provides high-quality complete protein, meaningful amounts of riboflavin, pantoic acid, biotin, phosphorus, iron and zinc, together with some of the richest natural sources of choline, all within a single, compact food.

Carbohydrates and fats

The accompanying fats and carbohydrates also determine how a food fits into a balanced diet. Per 100g, boiled eggs contain 16.6 g of fat and virtually no carbohydrates, making them naturally suited to diets that prioritise high-quality protein. Cooked soybeans provide 7.9 g of fat, 1.7 g of carbohydrates, and an additional 792 g of dietary fibre, offering a combination of plant protein, complex carbs and fibre that supports sustained energy release.

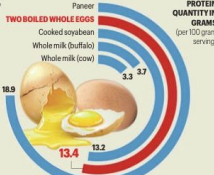
For the same quantity, paneer contains the highest fat content (14.8 g) among the foods compared, together with 12.4 g of carbohydrates, reflecting its energy-dense nature. Milk, meanwhile, remains the lightest option, contributing 4.4 g of fat and 49.8 g of carbohydrates, depending on the fat content.

These differences illustrate that while eggs stand out for their complete protein and micronutrient density, soybeans contribute valuable fibre and plant nutrients, and paneer offers a rich dairy nutrition, and milk provides a balanced everyday source of energy and nutrients. Each food serves a distinct nutritional purpose rather than competing on a single parameter. But from the lens of nutrition science, the humble egg continues to hold its place as one of the most complete and easy-to-prepare foods.

• EGGS vs THE REST

The chart and table show the quantity of nutrients such as protein and carbohydrates in various foods.

- Eggs and cooked soybeans appear indistinguishable in protein content. But not all proteins are created equal.
- Eggs are highly digestible and 'bio-available' (well-utilised by the body).
- Soybeans is a high-quality plant protein. But it is best as part of a dietary mix rather than a standalone solution.
- Paneer has higher protein. But 100g is tough to consume at one go. Egg also contains all nine essential amino acids.



NUTRIENT VALUES (BELOW ARE PER 100 GRAM SERVING)					
Nutrient	Whole milk (buffalo)	Whole milk (cow)	Paneer (cottage cheese)	Cooked soybean	Two boiled whole eggs
Energy (gram)	107.3	72.9	257.9	132.09	142.7
Carbohydrate (g)	8.3	4.9	12.4	3.57	Virtually nil
Iron (g)	6.5	4.5	14.8	6.79	10.5.4
Calcium (milligram)	121	118	476	68.25	55.1
Riboflavin (B2) (mg)	0.13	0.11	0.1	0.08	0.19

DATA COMPILED BY DR. SEEMA GULATI

A lesson from Tamil Nadu, pioneer in midday meals



ANGHA JAYAKUMAR

THE WEST Bengal government has announced that it will remove eggs from midday meals, sparking a debate over what children should be served in school.

The decision has prompted comparisons with Tamil Nadu, which, as Madras Presidency, pioneered the concept of midday meals in India, and has provided eggs under its programme for decades now. Tamil Nadu's midday meal scheme has evolved over century, with governments reviving it for children should be fed, and why.

Getting children into school...

Chief minister K Kamaraj is often credited with introducing a school meal programme in Madras State in 1956, but he had built on decades of experiments within the pre-independence Madras Presidency to get children into schools.

The first publicly funded school meal experiment in Madras Presidency began in 1920 as part of an effort to expand compulsory elementary education.

Officials recognised that poverty, hunger and families' dependence on children's earnings kept many children out of school. They also realised that compulsory education could not be achieved through coercive tactics alone. Against this backdrop, the Madras Corporation tried to school meal as a centre to attract and retain children in classrooms, when historian Caritona Ellis in her 2023 book *Imagining Childhood, Improving Children: The Emergence of an 'Auntie's' Role in Late Colonial South India*.

By most accounts, Justice Party president P. Theagaraya Chetty, also president of Madras Corporation Council, backed a midday meal scheme to benefit the state's impoverished children. The pilot was at a corporation school in Madras city in 1920.

The daily cost per head was capped at one anna. The scheme quickly proved its value. In *Imagining Childhood, Improving Children*, Ellis writes that attendance reportedly dropped by 50% when the scheme was briefly suspended in 1922.

By the end of the decade, the debate had

How school lunch came to other states

- In 1964, Kerala became the country's second state to have a school lunch programme.
- Over the next few years, many other states launched their own versions of the scheme, and finally in 1995, the Centre stepped in. It was launched as a centrally sponsored scheme across 2,408 blocks for the states up to Class 5, initially providing 50 grams of grains to states.
- In 2001, following a landmark Supreme Court order, states were directed to provide cooked mid-day meals rather than dry rations. In 2007, the UP government expanded the scheme to Class 8.
- In 2021, the scheme was named PM POSHAN.

shifted from getting children into school to ensuring they could learn once they were there. In 1940, the Madras Corporation declared, "If you cannot feed the body of a child, you cannot feed the brain." For Ellis, that shift was rooted in education rather than public health. "Hungry children can't concentrate on learning; they might be in school but they can't learn anything if they're hungry," she told *The Indian Express*.

...and keeping them there

By the 1950s, education authorities in Madras increasingly treated children's health as central to their educational mission. Ellis writes, "Superintendents became increasingly focused on the nutritional value of school meals, the need for a balanced diet and the quality of food provided to children. Officials debated whether bread and milk were preferable to the usual rice-and-curd diet, but also whether children should be served food as far as to them."

"The key conclusion was that it was more important to provide food that the children would like and eat, than to focus on the most efficient or nutritious food," Ellis told *The Indian Express*. "Providing food similar to what the children would have eaten at home seems to be a way of bridging school life with family and daily life, and therefore a place to learn."

Medical inspection increasingly accompanied school meals, reinforcing the idea that schools were responsible not only for education but also for nutrition and physical development. This marked a wider shift in how the state viewed children.

However, this political concern did not always translate into adequate funding.

Eggs in the basket

The next milestone came after Independence, when chief minister Kamaraj expanded the school meal programme in 1956. The scheme was transformed under chief minister MG Ramachandran's Nutrition Meal Programme in 1982, which placed greater emphasis on nutritional needs alongside improving attendance.

In January 1989, chief minister M. Karunanidhi announced that children under the scheme would receive one boiled egg every two weeks. Successive governments expanded the frequency of egg distribution and the menu itself. In 2003, Chief Minister J. Jayalalitha introduced the Variety Meals scheme to expand the nutritional value and appeal of the midday meal programme.

Why Eggs Matter for Schoolkids: The Midday Meal Debate

- **Nutritional Case for Eggs**

- Eggs are a complete protein source containing all nine essential amino acids; highly bioavailable and easy to prepare compared with most plant proteins

- **The West Bengal Decision**

- State government announced removal of eggs from midday meals, reviving the debate on what food should be served to schoolchildren

- **Tamil Nadu : Pioneer of Midday Meals**

- Madras Presidency began school lunch experiments in 1920; the scheme scaled up nationally by 1995 following Supreme Court directions

- **National Evolution of the Scheme**

- Kerala became the second state in 1984; a centrally sponsored scheme launched in 1995 and was renamed PM POSHAN in 2021

Why Eggs Matter for Schoolkids: The Midday Meal Debate

Mid Day Meal Scheme

It is the largest school feeding programme of its kind in the world, covering students enrolled in government schools from Classes 1 to 8. The basic objective of this scheme is to enhance enrolment in schools.

Nodal Ministry: Ministry of Education.

The programme was first introduced in 1925 for disadvantaged children in Madras Municipal Corporation.

It is not just a scheme, but a legal entitlement of all school-going children in primary and upper primary classes, through the National Food Security Act (NFSA), 2013.

The central and state governments share the cost of the Midday Meal Scheme, with the centre providing 60 percent and the states 40 percent.

A Unified Policy Architecture for India's Energy Future

A unified policy architecture for India's energy future

India has made remarkable progress in transforming its energy landscape over the past decade. From achieving near-universal household electrification and expanding access to clean cooking fuel to becoming one of the world's fastest-growing renewable energy markets, it has demonstrated a strong commitment to ensuring energy access while advancing sustainable development.

As India looks ahead to the goals of energy self-reliance by 2047 and net-zero emissions by 2070, the next phase of the energy transition will require an increasingly integrated approach to planning and governance. A policy brief released by the Indian National Science Academy (INSA) in May 2026 highlights the importance of a unified national energy framework that can help align diverse energy resources, technologies and institutions towards common national objectives.

The complexity of India's energy system

The need for such an approach is evident from the scale and complexity of India's energy system. While domestic energy production continues to expand, there is a dependence on imports for a significant share of oil and natural gas requirements. At the same time, energy demand is expected to grow steadily as economic development, industrialisation and urbanisation continue. Managing these multiple priorities, energy security, affordability, sustainability and economic growth, requires coordinated planning across sectors and fuels.

India has already established strong foundations through initiatives such as the Saubhagya Scheme, the Pradhan Mantri Ujjwala Yojana, and ambitious renewable energy programmes. Renewable energy installed capacity has grown from approximately 40 GW in 2015 to approximately 260 GW by 2025, reflecting a determination to diversify the energy mix. As



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A policy brief from the INSA-Centre for Science, Technology, Innovation and Policy outlines a four-pillar framework

the energy ecosystem becomes more diverse, however, greater coordination among generation, transmission, storage, distribution and emerging technologies will become increasingly important.

How the framework works

The INSA policy brief proposes a framework built around four mutually reinforcing pillars: adequacy, access, affordability and appropriate sustainability.

First, adequacy focuses on ensuring reliable and diversified energy supplies through a balanced portfolio of conventional and emerging energy sources, supported by modern infrastructure, energy storage and digital technologies. The objective is to strengthen energy resilience while reducing long-term vulnerabilities.

Second, access emphasises reliable and equitable energy services for all citizens. Building on the country's achievements in electrification and clean cooking access, the framework advocates strengthening last-mile delivery, improving service quality and expanding decentralised energy solutions where appropriate.

Third, affordability recognises that a successful energy transition must remain economically viable for households, businesses and industries. The framework highlights the role of innovative financing mechanisms, efficient markets and consumer-focused safeguards in supporting an inclusive transition.

The fourth pillar, appropriate sustainability, underscores the importance of pursuing sustainability in a manner that is aligned with India's developmental priorities and resource endowments. Rather than adopting a one-size-fits-all approach, the framework advocates solutions that reflect India's unique social, economic and environmental context.

This includes support for local communities, workforce development and region-specific transition pathways.

The policy brief also identifies circular economy practices and carbon capture, utilisation and storage (CCUS) as important cross-cutting enablers that can complement renewable energy deployment and contribute to reducing emissions from industrial sectors.

Recognising that energy transitions occur over decades, the framework proposes a phased approach. Near-term priorities include strengthening infrastructure, accelerating renewable energy deployment, supporting emerging technologies such as green hydrogen, and developing institutional mechanisms that can facilitate long-term coordination. Over time, the emphasis would shift toward deeper integration of low-carbon technologies, expanded use of bio-resources and the development of a more interconnected and resilient energy ecosystem.

Viewing energy as whole

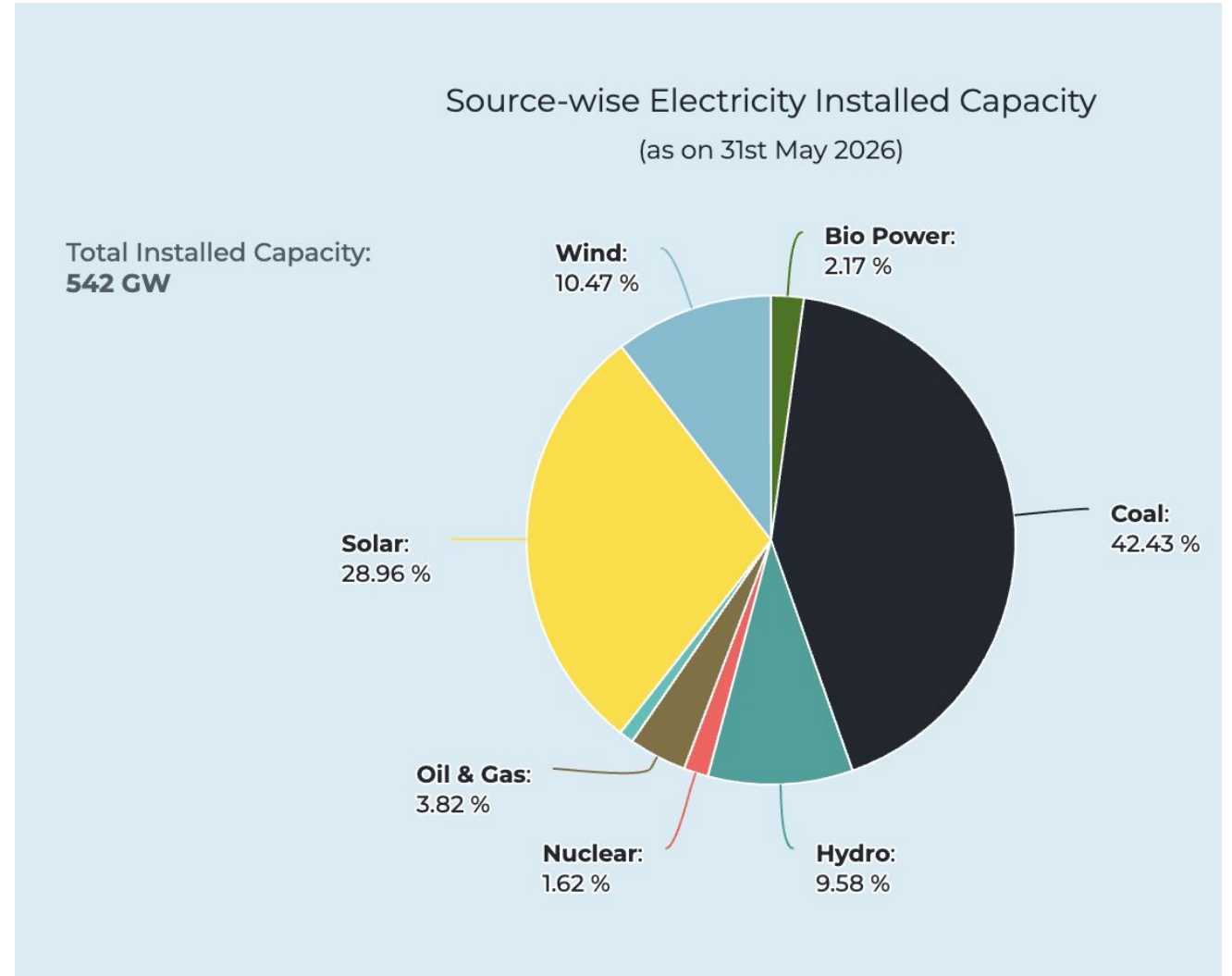
At its core, the framework highlights the value of viewing India's energy system as an integrated whole. Coal, renewables, biomass, natural gas, waste-to-energy systems and emerging clean technologies each have a role to play in supporting the country's development aspirations. Their effectiveness can be enhanced through greater coordination and long-term strategic planning.

India's energy transition is not only about expanding capacity; it is about creating a resilient, affordable and sustainable energy system capable of supporting future growth. By providing a common framework for aligning diverse energy pathways, the proposed approach offers a constructive road map for advancing national priorities while strengthening energy security for generations to come.

A Unified Policy Architecture for India's Energy Future

Installed Capacity

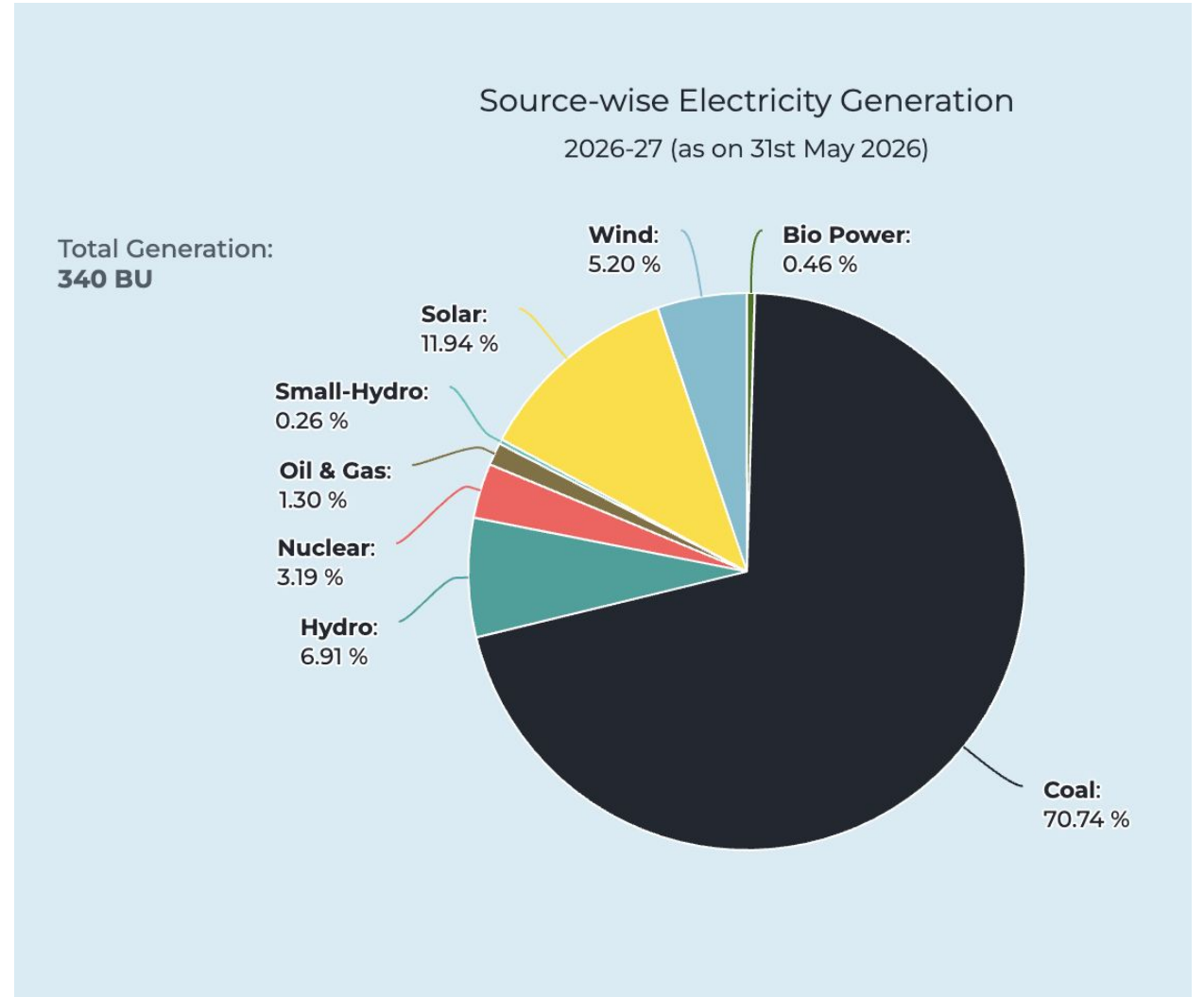
Coal > Solar > Wind > Hydro > Oil and Gas



A Unified Policy Architecture for India's Energy Future

Electricity Generation

Coal > Solar > Hydro > Wind > Nuclear



A Unified Policy Architecture for India's Energy Future

The National Statistics Office (NSO), Ministry of Statistics and Programme Implementation has released the annual publication “Energy Statistics India 2025”

Demand: It is driven primarily by industrial growth (40%), transportation (25%), and residential consumption (20%).

Energy Mix (Sources and Shares):

Coal: 48%

Oil: 28%

Natural Gas: 8%

Renewables (Solar, Wind, Hydro, Biomass): 12%

Nuclear: 4%

Coal Reserves and Production: The distribution of coal reserves in India is concentrated in a few states like Odisha (25.47%), Jharkhand (23.58%), Chhattisgarh (21.23%), West Bengal (8.72%) and Madhya Pradesh (8.43%).

A Unified Policy Architecture for India's Energy Future

Crude Oil: Geographically, the maximum crude oil reserves in India are concentrated in the Western Offshore region (32% of the total crude oil reserves). It is followed by the Assam region (22% of the country's crude oil reserves).

Natural Gas Reserves: Largest reserves of natural gas in India are located in the Western Offshore region (approximately 31% of the total natural gas reserves). It is followed by the Eastern Offshore (approx 24% of the reserves).

A Unified Policy Architecture for India's Energy Future

- **India's Energy Transition Progress**

- Near-universal household electrification and clean cooking access; renewable capacity grew from about 40 GW in 2015 to about 260 GW in 2025

- **INSA Policy Brief (May 2026)**

- Released by the Indian National Science Academy; proposes a unified national energy framework aligning diverse resources, technologies and institutions

- **Four Pillars of the Framework**

- **Adequacy (reliable, diversified supply), Access (equitable last-mile delivery), Affordability (economically viable transition), Appropriate Sustainability (context-specific, not one-size-fits-all)**

- **Cross-Cutting Enablers**

- Circular economy practices and Carbon Capture, Utilisation and Storage (CCUS) identified as key complements to renewable energy deployment

- **Phased, Long-Term Approach**

- Near-term focus on infrastructure and renewable deployment; long-term shift toward green hydrogen, bio-resources and deeper low-carbon integration toward net-zero by 2070

The Case for Building India's Coal Chemistry Capability

The case for building India's coal chemistry capability

There are two ways a country can survive an energy shock: by managing it skillfully through diplomacy, diversification, and fiscal measures, or by reducing dependence on the disrupted resource. India excelled at the first during the disruption in the Strait of Hormuz in 2026, with its refineries demonstrating exceptional technical flexibility in adapting to crude supply disruptions. The crisis reaffirmed that indigenous scientific capability and technological self-reliance are the decisive forms of insurance against energy market volatility – far more durable than any diplomatic or military arrangement alone. However, India has yet to reduce its underlying dependence, and coal offers a key opportunity to begin.

Same discipline for coal chemistry

Before turning to that opportunity, it is worth understanding why refinery flexibility proved so effective, because the same discipline will be required for coal chemistry. India's supplier base has nearly tripled over the past two decades. Each supplier provides a different crude slate, with distinct density profiles, sulphur content, and viscosity characteristics, and a refinery engineered for only one crude type becomes vulnerable to supply disruptions. Through investments in indigenous research, metallurgical advances, process innovation, and workforce training, India's refining sector developed the capability to process feedstock across a broad range of specifications. When the Strait of Hormuz closed and sourcing options shifted abruptly, Indian refineries adapted with technical confidence, processing crude from the Americas, the Atlantic Basin, West Africa, Russia, and India's West Asia partners. That flexibility at scale is the product of indigenous research and development, technical discipline, and engineers who understand their systems as interconnected processes rather than fixed machines.

The speed of the transition provides concrete evidence of this capability. Within weeks of the closure, non-Hormuz sourcing increased from 55% to 70% of India's crude intake. That pivot reflected a decade of upstream diversification combined with the downstream technical flexibility built into India's refinery fleet. India's private and public sector refineries had the engineering capability to process multiple crude types, adjust operating parameters at short notice, optimise fractionation patterns for different feedstock specifications, and maintain product quality and safety throughout the transition.

This capability was built through sustained investment in process understanding, operator training, and the institutional knowledge that enables a complex industrial system to absorb shocks without fracturing.

The liquefied petroleum gas (LPG) story offers a clear example of how indigenous refining



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capability can absorb a supply shock faster than markets can price it. India's LPG import infrastructure had roughly doubled over the preceding decade, providing greater distribution redundancy.

When the Strait of Hormuz closure threatened LPG availability, the bottleneck was not at the import ports but in how much LPG the existing refinery fleet could produce from the available feedstock. Under the LPG control order, refineries were directed to maximise yields, and within five days, domestic production increased from 35 Thousand Metric Tonnes (TMT) per day to 54 TMT per day, with engineers adjusting fractionation and cracking units in real time. That increase was engineering in action, not an accounting adjustment. It was one half of how India closed the gap; disciplined demand management provided the other. The production side – which is the focus of this article – rested entirely on technical capability built through years of sustained investment.

Energy security through molecules

Refinery flexibility solved the problem that the Strait of Hormuz crisis actually presented: how to keep a wide range of crude flowing through a fixed set of plants. It did not, and could not, solve the deeper structural problem the crisis exposed – that India's LPG dependence is far more concentrated than its crude dependence. A refinery can be engineered to process crude from 40 different countries. LPG, however, cannot be engineered to come from 40 different geographies, because the molecule is overwhelmingly sourced from a handful of Gulf and Atlantic Basin producers. The real long-term solution to LPG vulnerability is not refining the same imported molecule more efficiently. It is producing a domestic molecule that serves the same purpose.

That molecule already exists, and India has the raw material to produce it in extraordinary abundance. Dimethyl ether (DME) is a clean-burning gas chemically similar enough to LPG that it can be blended directly into existing cylinders and pipelines, requiring no new distribution network. It can be produced through coal gasification, which converts coal into syngas and then into DME. India possesses some of the world's largest coal reserves, and the Bureau of Indian Standards has already approved blending up to 20% DME with LPG. One recent industry assessment found that a 20% blend sourced from coal gasification could displace roughly 6.3 million tonnes of LPG imports each year, saving nearly ₹34,000 crore in foreign exchange annually. That is not a capital gain. It is the kind of structural reduction in import dependence that the Hormuz crisis should have taught India to take seriously.

This crisis has demonstrated how India's investments – in institutions, infrastructure, diplomacy and human capability – can translate

into national resilience. Innovation is often equated with breakthrough technologies. In reality, it is equally about creating new ways of integrating people, institutions, and ideas to solve unprecedented problems. The Ministry of Petroleum and Natural Gas's response exemplified this broader understanding of innovation.

From innovation to execution

Years ago, scientists at the CSIR's National Chemical Laboratory developed an indigenous technology for converting methanol into DME, a clean substitute for LPG. During the recent crisis, it was deeply gratifying to see the Centre for High Technology under the Ministry of Petroleum and Natural Gas move with remarkable speed to approve the scaling up of this indigenous pilot technology. It was a powerful reminder that investments in science made years earlier can become strategic national assets when unexpected crises arise.

This is exactly how innovation ecosystems should function. Research laboratories generate knowledge, government institutions identify strategic opportunities, and industry scales promising technologies. Together, they build national resilience.

That structural reduction is no longer waiting on policy. The Union Cabinet has approved a ₹37,500 crore scheme to promote surface coal and lignite gasification, explicitly citing the West Asia crisis as part of its rationale and targeting 100 million tonnes of coal gasification annually by 2030. The scheme provides an incentive of up to 20% of plant and machinery costs, separate from the DME blending ratio discussed above, and extends coal linkage tenure to 30 years – the kind of long-term horizon certainty that capital-intensive process industries require before committing investment. What remains is execution. India's coal has a higher ash content than the cleaner coal that underpinned China's dominant coal-to-chemicals industry, and domestic gasification capacity is still far below the ambition this scheme represents. Closing that gap is now a question of industrial discipline and investment, not policy intent. The intent has already been settled.

The remaining work – closing the ash-content gap, scaling gasification capacity, and building the technical depth China has spent two decades accumulating – is the same kind of work India's refining sector undertook over two decades of investment in metallurgy, catalysis, and process engineering. The lesson of Hormuz is not that India's refineries were ready and nothing else needs to change. It is that indigenous capability, once built, becomes a permanent strategic asset, and that the policy commitment to building the next one is now in place. The molecule is different, but the discipline required to master it is exactly the same as that which built the refineries that carried India through this crisis.

India's refinery resilience in the West Asia crisis points to its coal chemistry future

The Case for Building India's Coal Chemistry Capability

- **The 2026 Strait of Hormuz Crisis**

- Crude and LPG supply disruptions tested India's energy resilience; refineries adapted quickly by diversifying crude sourcing

- **Refinery Flexibility as Insurance**

- Non-Hormuz crude sourcing rose from 55% to 70% within weeks, built on two decades of indigenous refining R&D and a diversified supplier base

- **The LPG Vulnerability**

- Unlike crude, LPG is sourced overwhelmingly from a handful of Gulf and Atlantic Basin producers, making it structurally harder to diversify

- **Dimethyl Ether (DME) as a Solution**

- A clean LPG substitute produced via coal gasification; BIS has approved blending up to 20% DME with LPG, potentially saving about 6.3 million tonnes of LPG imports annually

- **Policy Push**

- Union Cabinet approved a Rs 37,500 crore scheme to promote coal and lignite gasification, targeting 100 million tonnes annually by 2030

On Curbing Young Adults on Social Media

On curbing young adults on social media

Incumbent Prime Minister of the U.K. Keir Starmer recently announced a policy to ban social media for children under-16. This was part of an arc of initiatives and announcements in Australia, Indonesia, Malaysia, France, and Canada to curb children's access to social media. This has triggered debates in India, especially at the State level, along the same lines.

The various contentions against a social media ban are not uniformly convincing. There is the argument that bans limit an "important source of information and learning" for children. However, the extent to which social media is used by children for learning is empirically arguable. It is also normatively questionable whether such platforms are an apt space for learning in the first place.

Scientific literature is full of intense debates regarding a clear and uniform correspondence between children's usage of social media and harmful experiences. Young adults respond to similar online circumstances differently; more so in jurisdictions that are as socially, economically, and digitally variegated as ours. What is required, foremost, is to identify the conditions where a high degree of correspondence does exist, and understand who are the most vulnerable among the expansive category of 'children'.

Ineffective bans

Meanwhile, there are numerous reasons why a ban for below-16s is unlikely to be effective. Curtailing access assumes that stout mechanisms of age verification exist on platforms. One must also be aware that age-verification allows social media platforms to collect sensitive identity data of under-16s. Bans tend to get circumvented by teenagers using the credentials of a willing family member or older peers. Even without external support, many young adults can manage to gain access though technological circumvention. There is also the



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Given the limited efficacy of bans, the alternative is to focus on the governance of social media platforms. This moves attention from those at risk to those contributing to creating risk

anxiety that bans could inculcate a wider culture of work-arounds among young adults – a behaviour which once ingrained could extend, in the years to come, to offline legal responsibilities too.

A combination of slack age verification systems and technological workarounds makes a mockery of a ban. Besides, the Australian experience has shown that 'age-gating' on social media platforms has pushed under-16s to migrate to less entrenched services; we are yet to know whether they are safe spaces.

Given the limited efficacy of bans, the alternative is to focus on the governance of social media platforms. This moves attention from those at risk to those contributing to creating the conditions of risk. Discussions on young adults' safety offer a potent opportunity to deepen debates on platform governance in India.

Charitable arguments point at design defects in some social media platforms that drive cravings and dependence among under-16s. But let's face it – addiction is central to the aims and operations of all service providers in the converged media environment. In fact, in the attention economy, platforms cannot compete without instituting, by design or deceit, some forms of dependence.

If addiction is the key concern, the Chinese response of capping screen time for children seems better directed. It amounts to the state mandating platforms to do what many concerned and informed parents do through other means. That said, in countries relatively less regulated, surveilled, and socially regimented than China, practices of borrowing credentials and technological workarounds to surmount screen-time caps will continue. Moreover, like bans, instituting caps requires age-verification, and will thereby legitimise platforms to harvest additional data of under-16s.

Limiting access and regulating platforms involve different

registers of monitoring and enforcement. In the case of a ban or caps on social media use by under-16s, two levels of enforcement can be visualised. The first is at the individual and household level; here, enforcing a ban, if feasible at all, would lead to intrusive forms of monitoring by platforms. The second will be at the platform level; a ban or screen time caps would demand age-verification and safeguards to privacy by platforms be rigorously enforced by the government.

Holding platforms accountable

The alternative is to legally oblige platforms to be transparent on their design, and provisions on safe spaces. Here, governments are likely to face structural and operational barriers. Governments that rely on social media giants to promote or protect their own interests are unlikely to push platforms to create transparent and accountable mechanisms. Furthermore, ensuring compliance on matters of design would make governments vulnerable to allegations of selective enforcement.

Evidently, the policy option of curtailing access is easier than that of making platforms accountable. Governments find it socially acceptable and legislatively convenient to make a case for age-gating. Devising effective and fair protocols that hold platforms accountable for their unwillingness to design safe spaces is far more complex.

Those advocating to curtail access should be cognisant of three matters. Foremost, they ought to evidentially make their case against social media platforms. Second, they should realise that bans or capping screen-time is nowhere close to a well-rounded regulatory strategy to tackle online safety. Finally, they must anticipate the burden on regulatory and judicial bodies that bans and their enforcement could lead to, and the perils to privacy that come with age-verification.

On Curbing Young Adults on Social Media

- **The Global Trend**

- The UK, Australia, Indonesia, Malaysia, France and Canada have moved to ban or restrict social media access for children under 16

- **Why Bans May Be Ineffective**

- Weak age-verification systems, workarounds via borrowed credentials, and migration to less-regulated platforms undermine the intent of bans

- **The Governance Alternative**

- Shifts attention from restricting at-risk children to holding platforms accountable for design choices that drive addiction and dependence

- **China's Screen-Time Cap Model**

- **State-mandated caps on children's screen time are seen as more targeted than outright bans, though they still require age-verification**

Army Chief Unveils Road Map 'VIJAY' for a Future-Ready Force

Army chief unveils road map 'VIJAY' for a future-ready force

Saurabh Trivedi
NEW DELHI

Chief of the Army Staff General Dhiraj Seth on Wednesday unveiled a road map titled “VIJAY”, outlining his vision for transforming the Indian Army into a technology-enabled, future-ready force capable of operating across multiple domains. The road map, inspired by the Defence Minister’s vision for the Decade of Transformation, was unveiled a day after he assumed charge as the 31st Chief of the Army Staff.

Calling it a moment of “great pride and humility” to lead the Indian Army, General Seth reaffirmed his commitment to the principles of “Duty, Honour and Nation First” and thanked the Prime Minister and the Defence Minister for reposing their faith in him.

Describing the Indian Army as a combat-ready and battle-hardened force, the Army Chief said it re-



General Dhiraj Seth inspects the Guard of Honour, at the South Block Lawns, in New Delhi on Wednesday. SUSHIL KUMAR VERMA

mains fully prepared to meet emerging security challenges.

Explaining the road map, General Seth said “V” represents Vigilance, with emphasis on maintaining operational readiness and constant alertness along the borders. “I” stands for Innovation and Transformation, focusing on modern doctrine, cutting-edge technologies and capability development to keep pace with the changing character of warfare. The letter “J” signifies Jointness and Integration, highlighting deeper synergy among

the Army, Navy and Air Force, alongside greater military-civil fusion through a Whole-of-Nation approach to strengthen national security. For “A”, he emphasised Atmanirbharta, reiterating the Army’s commitment to indigenous technologies. “Y” stands for Yodha First, reflecting the Army’s focus on every soldier as its greatest strength. He said improving training standards, technological skills, welfare and the empowerment of veterans and Veer Naris would remain key priorities.

Army Chief Unveils Road Map 'VIJAY' for a Future-Ready Force

- **The VIJAY Road Map**

- Unveiled by General Dhiraj Seth, the 31st Chief of the Army Staff, a day after assuming charge; inspired by the Defence Minister's vision for the "Decade of Transformation"

- **Decoding VIJAY**

- V-Vigilance (operational readiness and border alertness), I-Innovation & Transformation, J-Jointness & Integration, A-Atmanirbharta, Y-Yodha First

- **Jointness & Military-Civil Fusion**

- Emphasis on deeper synergy among the Army, Navy and Air Force, alongside a "Whole-of-Nation" approach to strengthen national security

- **Focus on the Soldier**

- Priorities include improving training standards, technological skills, veteran welfare, and support for Veer Naris (war widows)

- **Strategic Context**

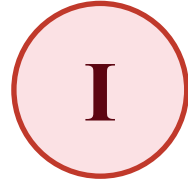
- Positions the Army as a combat-ready, technology-enabled force capable of operating across multiple domains

Decoding VIJAY



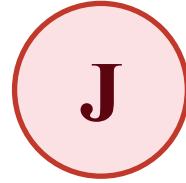
Vigilance

Operational readiness & border alertness



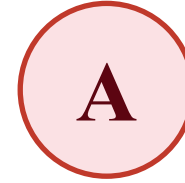
Innovation & Transformation

Modern doctrine, cutting-edge technology



Jointness & Integration

Synergy among Army, Navy, Air Force



Atmanirbharta

Commitment to indigenous technologies



Yodha First

Every soldier as the Army's greatest strength

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Practice Question

**The road map 'VIJAY', recently unveiled by the Chief of the Army Staff, stands for which of the following?
(PYQ-style)**

- (a) Vigilance, Innovation & Transformation, Jointness & Integration, Atmanirbharta, Yodha First**
- (b) Valour, Integration, Justice, Autonomy, Youth**
- (c) Vision, Infrastructure, Jointness, Alliance, Yield**
- (d) Vigilance, Indigenisation, Justice, Ability, Yodha**

Thank You